VisionSense

AI vision & colour-blindness screening

iQOO
HEALTHTECH
Round-1 Brief

■ **SENSOR:** Screen (optotypes) · Camera (distance) · Touch

The problem

People notice blurry vision only after it has worsened, and colour-blindness often goes undetected for years — affecting school, work and daily life. Professional eye tests aren't always convenient.

How it works

- Front camera estimates face distance so letters map to a true visual angle.
- Tumbling-C acuity staircase → logMAR / 6-6 estimate per eye.
- Ishihara-style plates screen colour vision (protan/deutan/tritan) with a severity estimate.
- Amsler grid checks central-vision distortion.
- Colour tools (namer + Daltonize filter) help colour-blind users in daily life.

Example output

Estimated distance acuity: 6/9 (logMAR 0.18)
Colour vision: likely deutan (red-green) — mild
Acuity decreased consistently across 3 tests — see an optometrist
Amsler grid: no distortion reported

Metrics

Acuity (logMAR / 6-6) · colour score & type · Amsler result · change over time

Novelty & positioning

Focused — one-shot acuity apps are mature (Peek, EyeQue); differentiator is change-monitoring + colour-blind assistive tools (Daltonize).

Target users

Students · glasses-wearers · children (myopia) · colour-blind users · low-access populations

Assistive / 'cure' feature

Doesn't cure — colour blindness is genetic (Daltonize + namer make life easier); refractive blur → optometrist; Amsler flag → specialist.

Research backing

- Smartphone acuity within <1 line of clinical charts (PMC 2023)
- Peek Acuity validated for screening, AUC >0.90 (Optometry 2024)
- Smartphone refractive-change monitoring feasible (ARVO 2026)

On-device AI coach	**Personal baseline + drift**	**History & charts**	**Monthly PDF report**	**Contact-a-specialist**	**100% private / offline**

Part of the **SenseKit** suite (MotionSense · SpeakSense · HearSense · VisionSense). Screening & self-monitoring only — not a medical diagnosis. Phase 2 = native app with calibrated sensors.